Local Income Taxes and Residential Sorting
Evidence from Pennsylvania LEHD Origin–Destination Data

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May 7, 2026

Motivation

- Tiebout (1956): citizens sort across jurisdictions according to their demands for local public goods.
- Strong evidence of tax-driven sorting at the state level (**??**).
- Little causal evidence on local income tax sorting within metro areas
- Two data problems make this hard. Local tax jurisdictions are sub-county, so IRS/ACS migration data are too coarse. Credit systems and weak enforcement complicate the effective tax burden.

This Paper

- Question: Do local income tax rate changes cause sorting?
- Setting: Pennsylvania, 2002–2019. Universal Earned Income Tax with continuous rate variation across ∼2,500 munis × ∼500 school districts.
- Data: LEHD LODES origin–destination, linking residence tract with workplace tract annually.
- Outcomes: `share_live_in` = (workers who live and work in city i) / (total workers in i), plus count outcomes (resident workers, total jobs).
- Exogeneous identification: Act 32 consolidated collection and made employer withholding mandatory.

Institutional Setting (Pennsylvania)

- Universal Earned Income Tax. About 2,500 munis and 500 school districts each set their own rate. Typical 0.5% muni + 0.5% school district.
- Before Act 32, collection was fragmented and withholding was inconsistent.
- Act 32 (effective 2012) consolidated collection and required employers to withhold the higher of the home and work rates.

How the PA EIT works (post-Act 32, excluding Philadelphia)

For a PA resident working outside Philadelphia, the applicable EIT rate is the higher of the worker's total resident rate and the work-location nonresident rate:

$$T_{ij} = \max\{\tau_j^{\text{res}} + \tau_j^{\text{SD}},\ \tau_i^{\text{nr}}\}.$$

Liability allocation:

$$\left(\tau^{\text{home}},\ \tau^{\text{work}}\right) = \begin{cases} \left(\tau_j^{\text{res}} + \tau_j^{\text{SD}} - \tau_i^{\text{nr}},\ \tau_i^{\text{nr}}\right) & \text{if } \tau_j^{\text{res}} + \tau_j^{\text{SD}} > \tau_i^{\text{nr}} \\ \left(0,\ \tau_i^{\text{nr}}\right) & \text{if } \tau_j^{\text{res}} + \tau_j^{\text{SD}} \leq \tau_i^{\text{nr}}. \end{cases}$$

- Work muni collects up to its nonres rate; excess (if any) goes to home muni and SD.
- Home jurisdictions get nothing if work nonres rate is higher.
- Binding rate determines the margin: resident binds ⇒ moving lowers bill; work binds ⇒ only job change does.

How the PA EIT worked before Act 32

Before Act 32, the underlying liability was still credit-based: workers owed the work-location nonresident tax, and the home jurisdictions received any residual after crediting work-location tax.

$$T_{ij}^{pre} = \tau_i^{\mathrm{nr}} + \max\left\{0,\ \tau_j^{\mathrm{res}} + \tau_j^{\mathrm{SD}} - \tau_i^{\mathrm{nr}}\right\} = \max\left\{\tau_j^{\mathrm{res}} + \tau_j^{\mathrm{SD}},\ \tau_i^{\mathrm{nr}}\right\}.$$

- Liability formula identical to post-Act-32. The change was *administrative*: fragmented local collection, weak employer withholding, no uniform residency certification.
- Act 32 standardized withholding, remittance, residency certification, collection districts, and revenue distribution.

Worked example

Worker 1 lives in city B and works in city A. Home muni rate is 1.0%, home school district is 0.5%, so the total home rate is 1.5%. The work-location nonresident rate is 0.8%.

- Worker 1 pays the higher of the two rates, so 1.5%. Home collects $1.5 - 0.8 = 0.7\%$; work collects 0.8%.
- This is a home-binds case. To lower the bill, worker 1 would have to move out of B. Changing jobs would not help because the work rate is already lower.

Philadelphia: a separate regime

- Philadelphia is exempt from Act 32. Its Wage Tax operates under the Sterling Act of 1932, administered by the City Department of Revenue.
- Philly residents pay the Philly resident rate regardless of where they work. There is no max rule and no offset.
- Nonresidents working in Philly pay the Philly nonresident rate, which goes entirely to Philly.
- In my analysis I keep Philly in the main sample because it is part of PA's labor market, but exclude it in robustness checks because its reporting and dynamics differ from Act-32 munis.

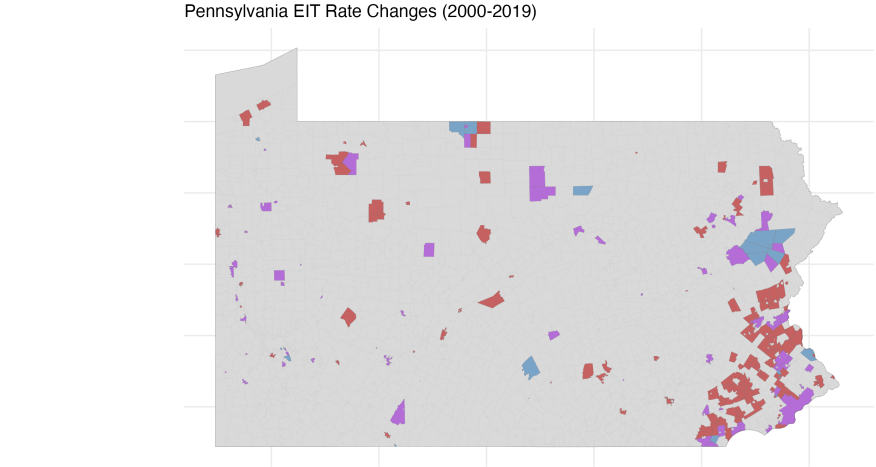
Data Generating Process

1. Pull Census tract LODES OD flows (LEHD, 2002–2019).
2. Scrape PA rates from DCED MunStats: muni resident, muni nonresident, and school district EIT for each muni-year.
3. Scrape PA municipal Annual Financial Reports (DCED) for spending on police, fire, streets, sewer, parks, comm dev, debt, and state aid.
4. Crosswalk DCED munis to Census (99.6% match) and aggregate tract-pair flows to muni-pair-year.

Summary Statistics (Pennsylvania, 2002–2019)

Variable	Mean	Median	SD	Max
share_live_in	0.103	0.088	0.076	0.657
Muni resident EIT (%)	0.532	0.500	0.175	4.331
Muni nonres EIT (%)	0.891	1.000	0.235	3.820
SD EIT (%)	0.617	0.500	0.289	2.050
Total jobs	2,438	565	14,502	629,698
Resident workers	450	50	8,001	391,047
Muni × year	35,743 (2,275 munis)			

Treatment Geography: PA Rate Changes



Red = increased; blue = decreased; purple = both; grey = unchanged.

Three Channels of Adjustment

When muni A raises its rate, residents of A can:

1. Move residence to a lower-rate muni, keep job in A. Saves at least the home SD rate at A.
2. Move both residence and job to a lower-rate muni. Largest savings.
3. New entrants pick a home outside A for a given job.

All three reduce `share_live_in` for A.

Q2: Does the rate increase cause sorting?

Workplace panel; TWFE & DCDH on `resident_eit`.

Outcomes

Unit i = PA municipality, t = year, 2002–2019.

- $\texttt{share_live_in}_{it} = \dfrac{\text{live + work in } i}{\text{total workers in } i}$
- $\log(\text{Resident workers}_{it})$ = workers who live in muni i, any work location.
- $\log(\text{Total jobs}_{it})$ = workers employed in muni i, any home location.

Q2: Do rate increases cause sorting?

Two specs

$$\text{Total stack:} \quad Y_{it} = \alpha_i + \gamma_t + \beta\, \tau_{it}^{\text{home}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

$$\text{Decomposition:} \quad Y_{it} = \alpha_i + \gamma_t + \beta^{m} \tau_{it}^{\text{res,muni}} + \beta^{sd} \tau_{it}^{\text{SD}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

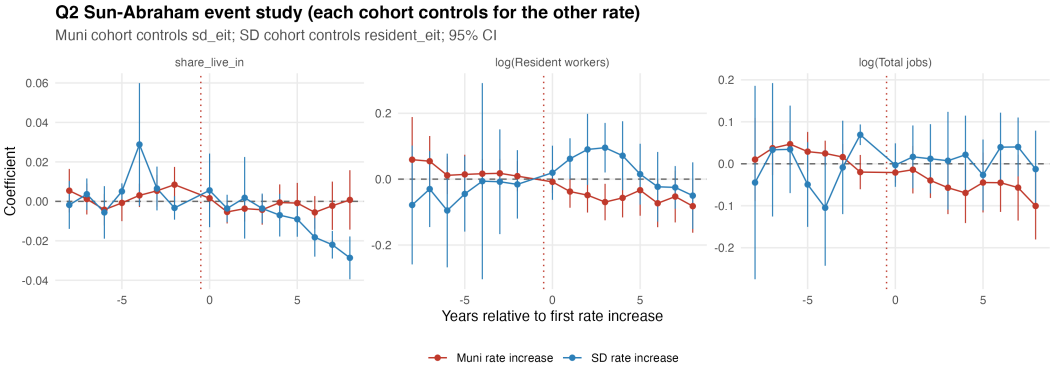
- Treated = munis that ever raised their muni or SD rate (within-city rate change drives identification).
- Control = munis whose rates stayed flat over the panel.
- $\mathbf{X}_{it}$ = log muni spending on police, streets, debt, state aid. Y_{it} = `share_live_in` or log count outcomes.

Q2 Answer: muni rate drives sorting; SD null

	share_live_in	log(Resident)	log(Total Jobs)
Total stack			
τ^{home}	-0.010^{**}	-0.137^{***}	-0.096^{**}
	(0.004)	(0.043)	(0.038)
Decomposition			
$\tau^{\text{res,muni}}$	-0.012^{**}	-0.201^{***}	-0.128^{***}
	(0.006)	(0.041)	(0.038)
τ^{SD}	-0.005	$+0.029$	-0.013
	(0.006)	(0.114)	(0.093)
N	12,891	12,781	12,891

- Controls: log muni spending + revenue shares (EIT, real estate).
- 1pp muni rate $\Rightarrow -1.2$ share-pp; -20% resident workers.
- SD null. Prior $+0.147^{**}$ on log(Resident) was a tract-centroid artifact (see Data note).

Q2 Event Study: muni vs SD decomposed



Q2 DCDH: CONTINUOUS TREATMENT, STAGGERED DESIGN

de Chaisemartin & D'Haultfœuille (2024) continuous estimator handles staggered treatment + continuous dose without TWFE's negative-weighting problem.

$$Y_{it} = \alpha_i + \gamma_t + \text{ACR} \cdot \tau_{it}^{\text{res,muni}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

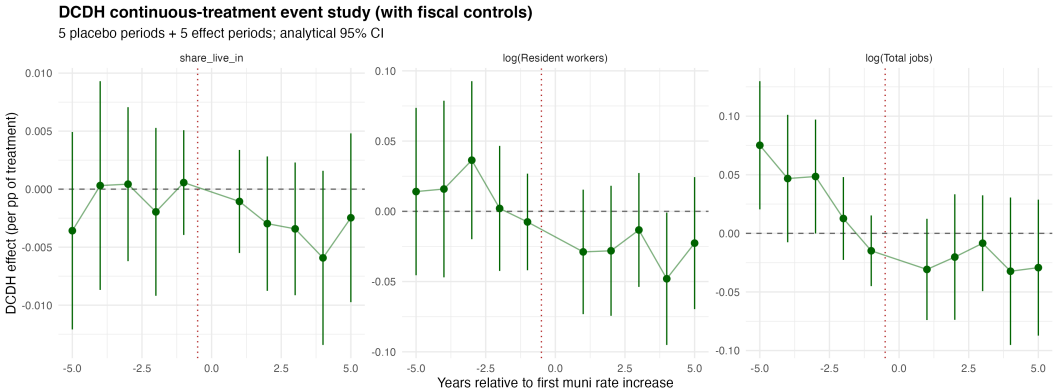
- Treated = munis ever raising muni rate.
- Control = munis whose rates stayed flat over the panel.
- Dose = muni resident rate (continuous).
- Effects up to $\ell = 5$; placebos up to $\ell = -5$.
- $\mathbf{X}_{it}$ = log muni spending on police, streets, debt, state aid. Y_{it} = `share_live_in` or log count outcomes.

Q2 DCDH Answer: average causal response per pp

	share_live_in	log(Resident)	log(Total Jobs)
ACR per pp	−0.009	−0.078	−0.067
	(0.006)	(0.047)	(0.055)
N (city-yrs)	12,926	12,926	12,926
Munis	948	948	948

- **Note: DCDH not yet rerun on the block-aggregated panel** (computationally intense). Numbers above are from prior tract-centroid panel; expected to shift modestly toward the new TWFE muni point estimates (-0.012^{**}, -0.201^{***}, -0.128^{***}).
- DCDH addresses the staggered-DiD critique: no negative weights from already-treated comparison groups.

Q2 DCDH Event Study



Mechanism: Is the effect enforcement-driven?

Act 32 split: do the same statutory rates produce different behavior pre- vs. post-2012 enforcement?

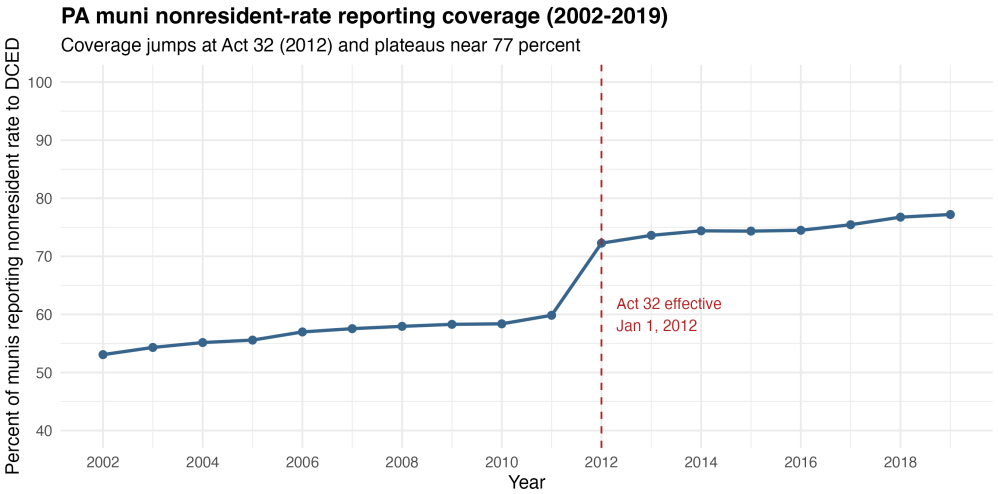
Mechanism: Is it enforcement? (Act 32)

Unit i = PA municipality; t = year. Outcome $\texttt{share_live_in}_{it}$. Structural break on rate effect:

$$Y_{it} = \alpha_i + \gamma_t + \beta^{\text{pre}} \tau_{it} \cdot \mathbf{1}[t < 2012] + \beta^{\text{post}} \tau_{it} \cdot \mathbf{1}[t \geq 2012] + \varepsilon_{it}$$

- Pre-2012: fragmented collection, weak withholding; DCED's nonresident rate registry was incomplete (only ∼55% of munis reported nonres rates; jumped to ∼72% in 2012).
- Post-2012: centralized, mandatory employer withholding at $\max(\tau^{\text{home}}, \tau^{\text{nr,work}})$.
- Run break for both rates separately to see which channel is enforcement-bound.

Nonresident rate reporting coverage



Do the rate increases we exploit actually bind?

Self-pair (worker lives + works in X): home stack $= \tau_X^{\text{res}} + \tau_X^{\text{SD}}$ vs. τ_X^{nr}.

- PA muni-years post-Act-32 (15,423): home binds 41%, work binds 59%; mean wedge +0.27pp.
- Muni-resident rate increases 2012–2019 (57 events): **84% (48/57) bind at home** ⇒ rate hike moves the bill.
- Remaining 16% (9/57): work binds already ⇒ hike is irrelevant.

Cross-pair within 80km, post-Act-32 (1.83M pair-years): work binds 61.6%; home binds 38.4%.

Implication: most increases are economically meaningful. Piecewise test below separates extensive switches from intensive within-channel changes.

Act 32 Answer: Q2 results split pre/post 2012

	share_live_in	log(Resident)	log(Total Jobs)
Decomposition			
$\tau^{\text{res,muni}} \times$ Pre	-0.008	-0.139^{***}	-0.073
$\tau^{\text{res,muni}} \times$ Post	-0.013^{**}	-0.209^{***}	-0.132^{***}
$\tau^{\text{SD}} \times$ Pre	-0.007	$+0.032$	$+0.027$
$\tau^{\text{SD}} \times$ Post	-0.003	$+0.026$	-0.038
N	12,891	12,781	12,891

- Controls: log spending (police, streets, debt, state aid) + revenue shares (EIT, real estate).
- Pre-Act-32 muni coefficient on share_live_in is null; post-Act-32 it is significant. Magnitudes on log(Resident) more than *double* from -0.139 pre to -0.209 post.
- SD coefficient is null in both periods — the prior $+0.147^{**}$ pre / $+0.170^{**}$ post pattern was an artifact of tract-centroid job misattribution.

Q3: Are increases and decreases symmetric?

Clean Inc-only / Dec-only / NC groups (drop flip-floppers).

Q3 Sample: clean treatment vs control groups

	Never-changed		Inc-only		Dec-only	
	Mean	SD	Mean	SD	Mean	SD
share_live_in	0.102	0.067	0.132	0.080	0.104	0.061
Muni resident EIT (%)	0.514	0.102	0.822	0.448	0.617	0.253
SD EIT (%)	0.590	0.253	0.556	0.214	0.483	0.090
Total EIT (%)	1.104	0.257	1.378	0.498	1.101	0.254
Total jobs	2,943	7,419	8,080	10,589	3,436	3,059
Resident workers	328	1,660	1,195	2,719	352	382
Munis	1,110		90		15	
Muni-years	11,076		1,126		178	

- Never-changed = munis with no rate change 2002–2019. Inc-only = ever raised, never decreased. Dec-only = ever decreased, never raised.
- Flip-floppers (raised AND decreased at some point) dropped.

Q3: Are rate changes symmetric? (clean groups)

Single regression: continuous muni rate interacted with group; SD rate enters as a single coefficient (no group interaction):

$$Y_{it} = \alpha_i + \gamma_t + \beta_g^m \tau_{it}^{\text{res,muni}} \cdot G_i + \beta^{sd} \tau_{it}^{\text{SD}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

where $G_i \in \{\text{nc, inc, dec}\}$ defined by muni-rate event.

- Treated: inc-only munis (90) and dec-only munis (15).
- Control: never-changed munis (1,110).
- Flip-floppers dropped.
- Continuous rate (per pp), not binary cohort.
- $\mathbf{X}_{it}$ = log muni spending (police, streets, debt, state aid) + revenue shares (EIT, real estate, intergov).
- $\tau^{\text{res,muni}}$: nc dropped (no within-variation by definition).

Q3 Answer: Inc effect robust; SD null

	share_live_in	log(Resident)	log(Total Jobs)
$\tau^{\text{res,muni}} \times$ Inc	-0.016^{**}	-0.221^{***}	-0.151^{***}
	(0.007)	(0.038)	(0.040)
$\tau^{\text{res,muni}} \times$ Dec	$+0.032^{**}$	-0.295	-0.067
	(0.016)	(0.657)	(0.114)
τ^{SD}	-0.004	$+0.024$	-0.018
	(0.006)	(0.122)	(0.099)
N	12,298	12,188	12,298

- Inc-only: -1.6 share-pp per pp; -22% resident workers; -15% total jobs.
- Dec (15 munis) noisy; $+0.032^{**}$ is small-sample composition, not reversal.
- SD null — prior $+0.139^{**}$ was tract-centroid artifact.

Q3 Full Results (all coefficients)

	share_live_in	log(Resident)	log(Total Jobs)
$\tau^{\text{res,muni}} \times$ Inc	-0.016^{**} (0.007)	-0.221^{***} (0.038)	-0.151^{***} (0.039)
$\tau^{\text{res,muni}} \times$ Dec	$+0.032^{**}$ (0.016)	-0.296 (0.657)	-0.067 (0.114)
τ^{SD}	-0.004 (0.006)	$+0.024$ (0.122)	-0.018 (0.099)
log_police	-0.000 (0.001)	$+0.004$ (0.012)	$+0.010$ (0.012)
log_streets	-0.001 (0.001)	$+0.010$ (0.010)	$+0.018^{**}$ (0.009)
log_debt	-0.000 (0.001)	-0.005 (0.005)	$+0.001$ (0.005)
log_state_aid	$+0.001$ (0.001)	$+0.011$ (0.010)	-0.005 (0.009)
share_eit	$+0.005$ (0.010)	-0.046 (0.118)	-0.105 (0.114)
share_re	-0.001 (0.008)	-0.076 (0.092)	-0.074 (0.088)
N	12,298	12,188	12,298

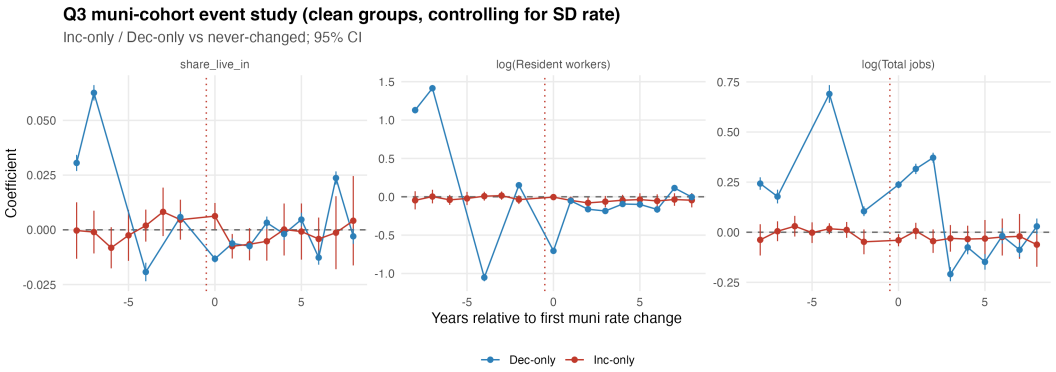
Cluster-robust SEs in parentheses. City + year FE.

Q3 Bonus: How SD rate changes track muni-event groups

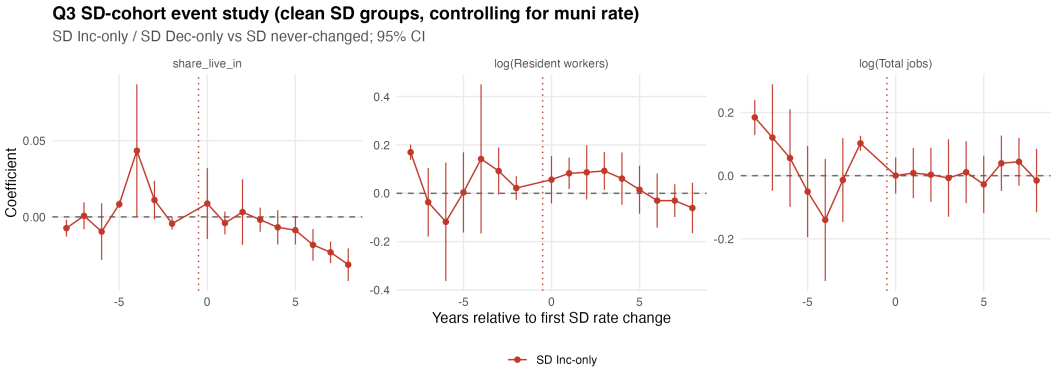
Muni group \ SD group	SD dec	SD flip	SD inc-only	SD nc
Muni dec-only (24 munis)	8%	0%	**50%**	42%
Muni inc-only (97 munis)	5%	0%	17%	78%
Muni nc (1,264 munis)	0%	0.4%	11%	89%

- Cities that cut their muni rate are *half* likely to be in an SD that raised its rate (50% vs 11% baseline).
- Cities that raise muni rates and never-changers mostly have stable SD rates (78–89% SD nc).
- Suggests a fiscal substitution pattern: SD raises rate → muni cuts to manage total burden, OR muni cut creates room for SD to raise. Act 1 of 2006 (PA SD rate referenda) was designed partly around this trade-off.

Q3 Event Study: Muni-cohort (Inc/Dec vs Never-changed)



Q3 Event Study: SD-cohort (Inc/Dec vs Never-changed)



SD inc-only ≈ 165 munis; SD dec-only too thin to fit.

Q3 Robustness: three experiments, four estimators

Block A (work $\times$ res rate) $\cdot$ Block B (work $\times$ nonres rate) $\cdot$ Block C (home $\times$ res rate)

Q3 Robustness motivation: staggered + multi-event treatment

- Of 113 inc-only munis, 17 (≈15%) raise rates more than once; 1 raises 9 times.
- Pooled TWFE on continuous `resident_eit` can have *negative weights* under heterogeneous treatment effects (Goodman-Bacon 2021, de Chaisemartin & D'Haultfoeuille 2020).
- Even after restricting to single-event munis, staggered timing (different munis treated in different years) keeps the problem alive — need SA / DCDH / stacked DiD.
- Four robustness checks below, each with both an aggregate coefficient and an event-study figure.

Q3 ROADMAP: THREE EXPERIMENTS, FOUR ESTIMATORS EACH

Q3 runs the same four estimators {(1) Single-event TWFE, (2) DCDH, (3) Stacked binary, (4) Stacked continuous} on *three* different (panel × rate) designs:

Block	Panel (unit)	Treatment	Outcomes
A	Workplace (work_city)	τ^{res} work-muni resident rate	share_live_in, log(Resident), log(Tot Jobs)
B	Workplace (work_city)	τ^{nr} work-muni nonres rate	share_nonres, log(NonRes), log(Tot Jobs)
C	Residence (home_city)	τ^{res} home-muni resident rate	share_work_outside, log(Residents), log(Work

- **A** = headline: do workers stop living where they work when the work-muni's resident rate rises?
- **B** = piecewise mechanism check: does the work-muni's nonresident rate push out cross-border workers?
- **C** = clean residential test: when home rate rises, do residents leave / shift to commuting out?
- Block A also includes a 5th estimator: piecewise binding model on cousub-pair flows.

(1) Single-event TWFE: drop multi-event Inc munis

$$Y_{it} = \alpha_i + \gamma_t + \beta\, \tau_{it}^{\text{res,muni}} + \delta\, \tau_{it}^{\text{SD}} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

Sample: 1,110 NC + 68 Inc-only with exactly one rate increase (drops 22 multi-event Inc munis). $\mathbf{X}_{it}$: log spending + revenue shares. SEs clustered at muni.

	share_live_in	log(Resident)	log(Total Jobs)
Total stack			
τ^{home}	−0.004 (0.005)	−0.062 (0.072)	−0.077 (0.061)
Decomposition			
$\tau^{\text{res,muni}}$	−0.004 (0.007)	−0.162*** (0.035)	−0.144*** (0.047)
τ^{SD}	−0.004 (0.006)	+0.027 (0.122)	−0.020 (0.099)
log_police	−0.000 (0.001)	+0.005 (0.012)	+0.011 (0.012)
log_streets	−0.001 (0.001)	+0.011 (0.010)	+0.018** (0.009)
log_debt	−0.000 (0.001)	−0.005 (0.005)	+0.001 (0.005)
log_state_aid	+0.001 (0.001)	+0.011 (0.010)	−0.005 (0.009)
share_eit	+0.003 (0.011)	−0.052 (0.125)	−0.104 (0.121)
share_re	−0.001 (0.008)	−0.076 (0.092)	−0.074 (0.088)
N	11,828	11,719	11,828

Cluster-robust SEs by muni in parentheses. City + year FE.

(1) Single-event sample ES (Sun-Abraham, binary cohort)

$$Y_{it} = \alpha_i + \gamma_t + \sum_{\ell \neq -1} \mu_\ell \, \mathbf{1}\{t - c_i = \ell\} + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{it}$$

SA cohort-aggregated (**?**); cohort c_i = first-increase year. Coefficients are average ATT in event-time, *not* per-pp dose.



68 single-event Inc vs 1,110 NC. Pre-trends near zero; post drift negative on log outcomes.

(2) DCDH: CONTINUOUS, MULTI-EVENT, INC-ONLY + NC

de Chaisemartin & D'Haultfoeuille (2024). For each pair of consecutive years ($t-1$, t) and each muni i, define the *instantaneous causal response*

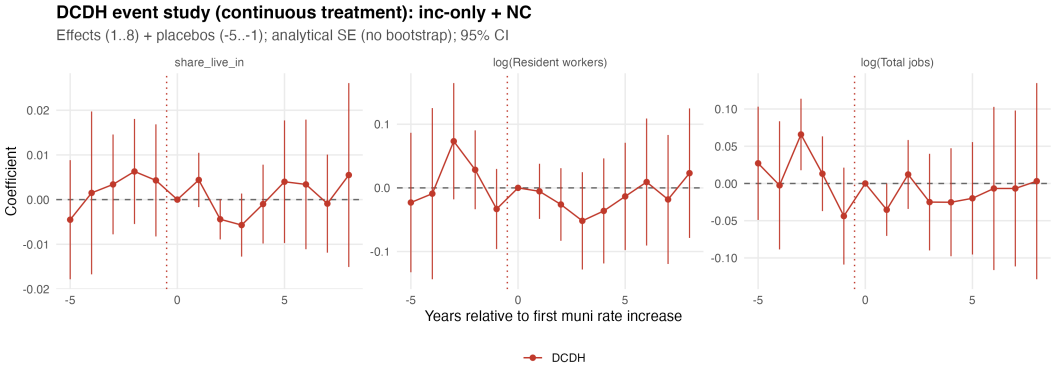
$$\delta_{i,t} = \frac{Y_{it} - Y_{i,t-1}}{\tau_{it} - \tau_{i,t-1}}$$

on switchers (munis with $\tau_{it} \neq \tau_{i,t-1}$). The estimator averages $\delta_{i,t}$ across switchers and event-times, using only never-treated munis as the difference-in-differences counterfactual; no negative weights even with continuous, staggered, multi-event treatment. ATE per pp:

	share_live_in	log(Resident)	log(Total Jobs)
Av. total effect	+0.002	−0.049	−0.043
	(−0.023, +0.026)	(−0.238, +0.139)	(−0.223, +0.137)
Effective N (cells)	10,853	10,750	10,853
Switchers	234	234	234

- ATE null but underpowered: only 36 switchers in t=1, falling to 21 by t=8.
- Analytical SEs are conservative under continuous treatment (package warns; bootstrap recommended for inference, ∼6–8 hr).
- Same controls as TWFE in the data setup; DCDH uses the residualized treatment after partialling controls.

(2) DCDH EVENT STUDY



Effects (years 1–8) + placebos (−5 to −1). Placebos not individually significant. Effect on log(Total jobs) at t=1 is −0.035 (95% CI −0.070, −0.000).

(3) Stacked DiD, binary post: 21 sub-experiments

For each cohort c (event year), construct a stack s: that treated muni plus all NC, $t \in [c-5, c+5]$. Concatenate stacks and estimate

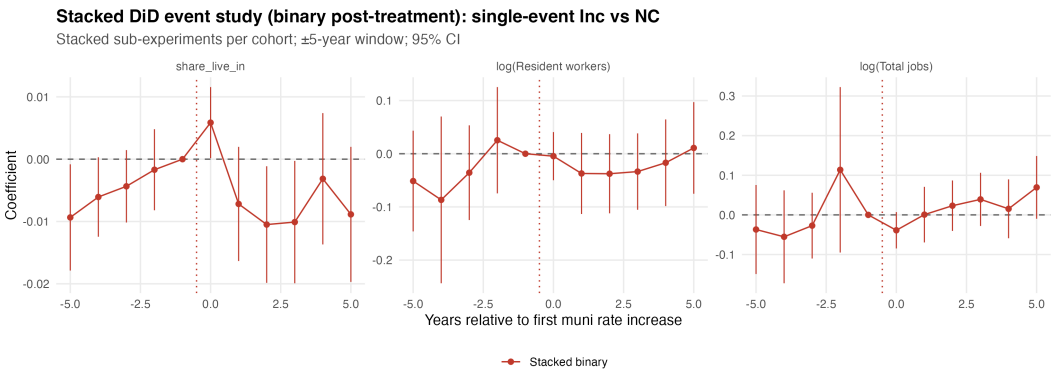
$$Y_{ist} = \alpha_{is} + \gamma_{ts} + \beta\,(\text{Treat}_i \cdot \text{Post}_{st}) + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{ist}$$

with stack×muni and stack×year FE. Only NC munis as controls ⇒ no already-treated comparison; SEs clustered at the original muni.

	share_live_in	log(Resident)	log(Total Jobs)
`treat_post`	−0.002 (0.003)	−0.000 (0.037)	+0.008 (0.029)
log_police, log_streets, log_debt, log_state_aid, share_eit, share_re	*(included; coefficients suppressed)*		
N (stack-muni-year)	95,287	94,317	95,287

- ATT pools all post-treatment years at intensive treatment = 1.
- Null because the intensity of the rate change is collapsed out — a 0.1pp hike and a 0.5pp hike both contribute equally.

(3) Stacked DiD binary event study



(4) Stacked DiD, continuous $\Delta\tau$: dose-response per pp

Same stacks as (3). Within each stack, define $\Delta\tau_{ist} = \tau_{it}^{\text{res,muni}} - \bar{\tau}_{is}^{\text{pre}}$ (rate deviation from the muni's own pre-treatment baseline within stack s). Then

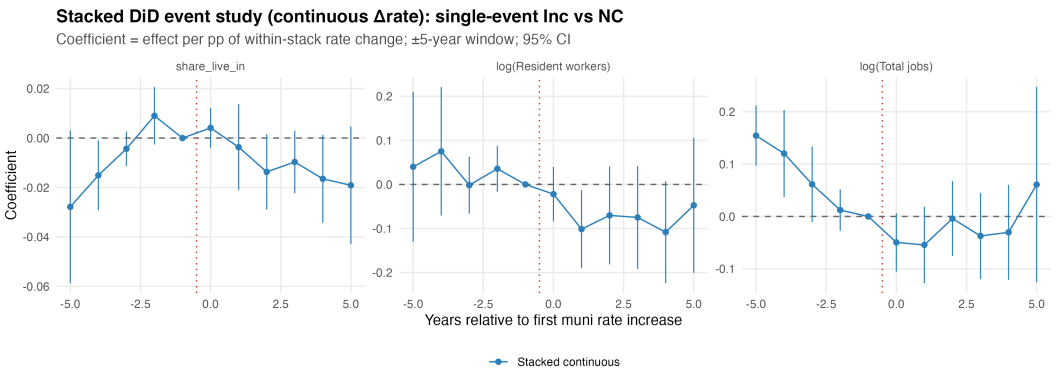
$$Y_{ist} = \alpha_{is} + \gamma_{ts} + \beta\,\Delta\tau_{ist} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{ist}$$

$\hat{\beta}$ has a per-pp interpretation. The event-study version interacts a time-invariant per-unit dose with $\mathbf{1}\{t - c_i = \ell\}$.

	share_live_in	log(Resident)	log(Total Jobs)
$\Delta\tau$ (per pp)	−0.005 (0.005)	−0.091** (0.036)	−0.072** (0.031)
log_police, log_streets, log_debt, log_state_aid, share_eit, share_re	*(included; coefficients suppressed)*		
N (stack-muni-year)	77,182	76,528	77,182

- **Recommended headline magnitude.** 1 pp muni rate increase ⇒ −9% resident workers, −7% total jobs.
- Roughly half the magnitude of Q2 TWFE pooled (−20%, −13%) — consistent with TWFE being inflated by negative weights under heterogeneous effects.
- No staggered-DiD critique: only NC munis as controls, no already-treated comparison.

(4) Stacked DiD continuous event study



Block A triangulation: Workplace × resident rate

Estimator	share_live_in	log(Resident)	log(Total Jobs)	Notes
Q2 TWFE (full)	-0.012^{**}	-0.201^{***}	-0.128^{***}	Pooled, possibly negative-weighted
Single-event TWFE	-0.004	-0.162^{***}	-0.144^{***}	Drops 22 multi-event Inc munis
DCDH ATE	$+0.002$	-0.049	-0.043	Null, wide CI; underpowered
Stacked binary	-0.002	-0.000	$+0.008$	Dose-information lost
Stacked continuous	-0.005	$\mathbf{-0.091}^{**}$	$\mathbf{-0.072}^{**}$	**Recommended headline**

- Direction is robust: *negative* on level outcomes across every clean estimator.
- Magnitude on log outcomes ranges from -0.04 (DCDH) to -0.20 (TWFE pooled). Stacked continuous -0.09 sits in the middle and has the cleanest identification properties.
- `share_live_in` is fragile across clean estimators; the level outcomes are the robust headline.

(5) Piecewise binding model: setup

Cousub-pair panel (work A, home B, year t). Under the institutional rule, the work jurisdiction always collects τ_A^{nr}; the home jurisdictions collect the *residual* $\max(\tau_B^{\text{home}} - \tau_A^{\text{nr}}, 0)$. Worker's total bill:

$$\tau^{\text{bill}}(A,B,t) \;=\; \max\!\big(\tau_B^{\text{home}},\; \tau_A^{\text{nr}}\big), \qquad \tau_B^{\text{home}} = \tau_B^{\text{res}} + \tau_B^{\text{SD}}.$$

Binding side = which jurisdiction's rate moves the bill at the margin: **home binds** iff $\tau_B^{\text{home}} > \tau_A^{\text{nr}}$, else **work binds**. For each pair and each year-over-year transition $(t-1)\to t$ (pooled across all $t \in$ 2013–2019), classify:

- **int_home**: home-binds in both $t-1$ and t.
- **int_work**: work-binds in both $t-1$ and t.
- **ext_to_home**: was work-binds, now home-binds.
- **ext_to_work**: was home-binds, now work-binds.

Decomposition of the year-over-year change in the bill:

$$\begin{aligned}\Delta \log \text{flow}_{ABt} = {} & \beta_{ih}\,\Delta\tau^{\text{bill}}_{ABt}\cdot\mathbf{1}\{\text{int_home}\} + \beta_{iw}\,\Delta\tau^{\text{bill}}_{ABt}\cdot\mathbf{1}\{\text{int_work}\} \\ & + \beta_{eh}\,\Delta\tau^{\text{bill}}_{ABt}\cdot\mathbf{1}\{\text{ext_to_home}\} + \beta_{ew}\,\Delta\tau^{\text{bill}}_{ABt}\cdot\mathbf{1}\{\text{ext_to_work}\} \\ & + \alpha_{AB} + \gamma_{At} + \varepsilon_{ABt}\end{aligned}$$

Each β is per-pp dose response within that channel-state. Pair FE α_{AB} + work-muni × year FE γ_{At} absorbs muni-A-specific year shocks (e.g., booming labor market that drives flows up while A raises rates). SE clustered at work muni.

(5) Piecewise: transition distribution post-Act-32

Pair-years 2012–2019, all 8.6M PA OD-pairs (no distance filter).

	STRICT (drop NA nonres)		*FILLED (NA → work res)*	
Transition	Pair-years	%	Pair-years	%
int_home	1,187,143	39.9	1,604,992	47.0
int_work	1,736,664	58.3	1,755,095	51.4
ext_to_home	19,521	0.7	20,446	0.6
ext_to_work	33,733	1.1	33,930	1.0
Total	2,977,061	100.0	3,414,463	100.0

- Extensive switches are rare (∼1.8% combined). Most pair-year action is intensive within a stable channel.
- Filling shifts pair-years toward home-binding (+7 pp) because imputed nonres = res ≤ res + SD = home stack wherever the imputation kicks in.

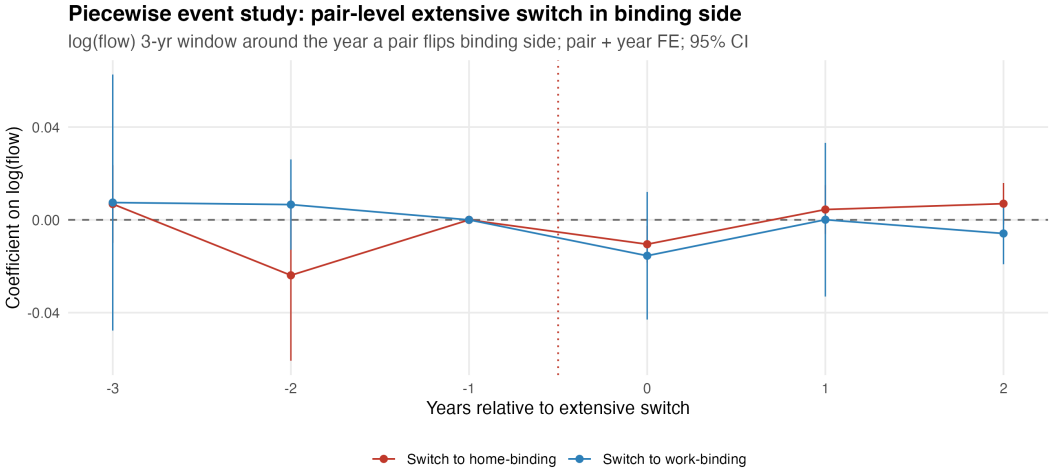
(5) Piecewise: aggregate and decomposed coefficients

log(flow), pair + year FE, SE clustered at work muni.

	STRICT		FILLED	
	Estimate	SE	Estimate	SE
Aggregate $\Delta\tau^{\text{bill}}$	+0.005	0.009	+0.001	0.008
β_{ih} (int_home)	+0.016	0.013	+0.001	0.012
β_{iw} (int_work)	+0.042	0.026	+0.043	0.027
β_{eh} (ext_to_home)	+0.002	0.012	+0.003	0.012
β_{ew} (ext_to_work)	+0.015	0.019	+0.016	0.019
N pair-years	2,735,968		3,129,492	
Pairs	823,244		953,578	

- All four channels null individually. Aggregate $\Delta\tau^{\text{bill}}$ null. Signs positive throughout, opposite the prediction.
- Pair + year FE absorb almost everything; signal of a $\sim$0.5 pp dose on a 1–3 job flow is too small to detect.
- $\beta_{iw} = +0.042$ ($p \approx 0.11$) wrong-signed — likely time-varying muni-A growth not absorbed by pair FE.

(5) Piecewise event study around extensive switches



76,119 pairs flip to home-binding; 52,570 flip to work-binding. Pre-trends near zero; post-flip drift toward zero is consistent with the null aggregate.

Block B (Workplace × nonres rate): motivation and group composition

Mirror of resident-side Q3 but treatment = τ_A^{nr} (work muni's nonresident rate); outcome = log(Nonresident workers) in muni A and log(Total Jobs).

- Re-classify munis on nonresident-rate events. Of 88 nonres-Inc munis, **83 had exactly one increase** (94% — multi-event scope is smaller than the resident side). Inc-only: 88; NC: 892; FF / Dec dropped.
- Direct test of the piecewise mechanism: nonresident workers face τ_A^{nr} when the work rate binds; resident workers don't. So the nonres-rate effect should concentrate on nonresident outflows / total jobs, not on resident workers.

(1) Single-event TWFE, nonres-side

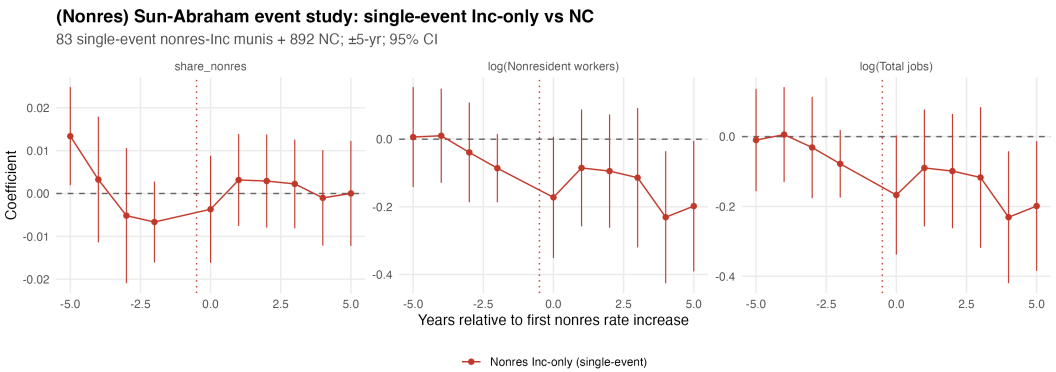
$$Y_{it} = \alpha_i + \gamma_t + \beta\, \tau^{\text{nr}}_{it} + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{it}$$

Sample: 892 NC + 83 single-event nonres-Inc munis.

	share_nonres	log(NonRes)	log(Total Jobs)
τ^{nr}	−0.004 (0.012)	−0.289* (0.169)	−0.286* (0.166)
N	9,911	9,911	9,911

Cluster-robust SEs by muni; same controls as resident side.

(1) Single-event ES, nonres-side (Sun-Abraham)



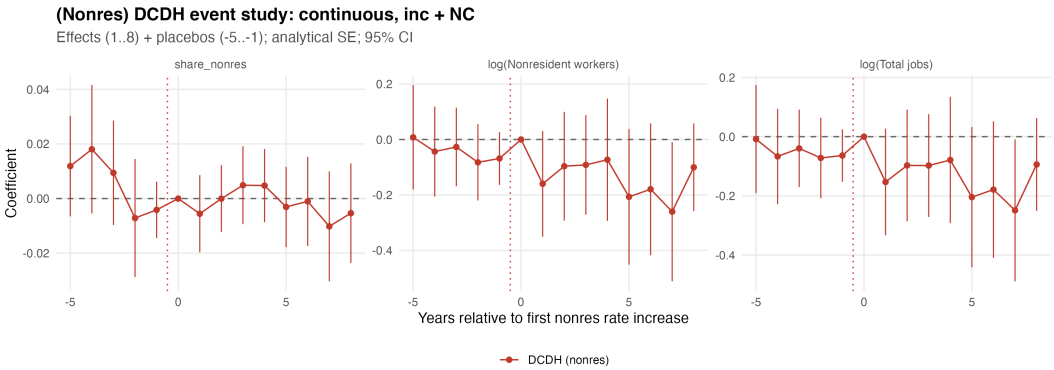
(2) DCDH, NONRES-SIDE: CONTINUOUS, MULTI-EVENT

Same DCDH machinery as resident side; treatment = τ^{nr}.

	share_nonres	log(NonRes)	log(Total Jobs)
Av. total effect	−0.004 (−0.028, +0.020)	−0.298 (−0.663, +0.068)	−0.294 (−0.646, +0.058)
Switchers	261	261	261
N (cells)	8,807	8,807	8,807

- Point estimates are similar to single-event TWFE (∼−30% on log NonRes and log Total Jobs) but CIs span zero. Bootstrap not yet run.

(2) DCDH ES, NONRES-SIDE



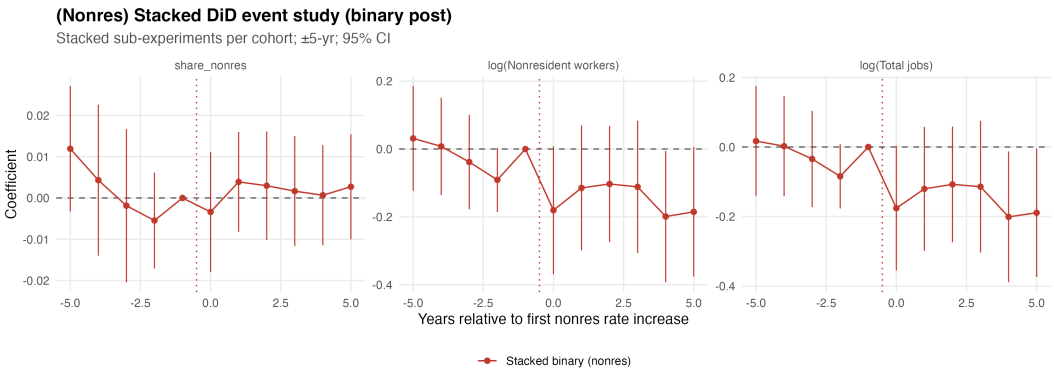
(3) Stacked DiD binary, nonres-side

$$Y_{ist} = \alpha_{is} + \gamma_{ts} + \beta\,(\text{Treat}_i \cdot \text{Post}_{st}) + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{ist}$$

18 sub-experiments (cohorts), ±5-yr window, NC-only controls.

	share_nonres	log(NonRes)	log(Total Jobs)
`treat_post`	0.000 (0.007)	−0.129 (0.086)	−0.130 (0.084)
N	76,292	76,292	76,292

(3) Stacked binary ES, nonres-side



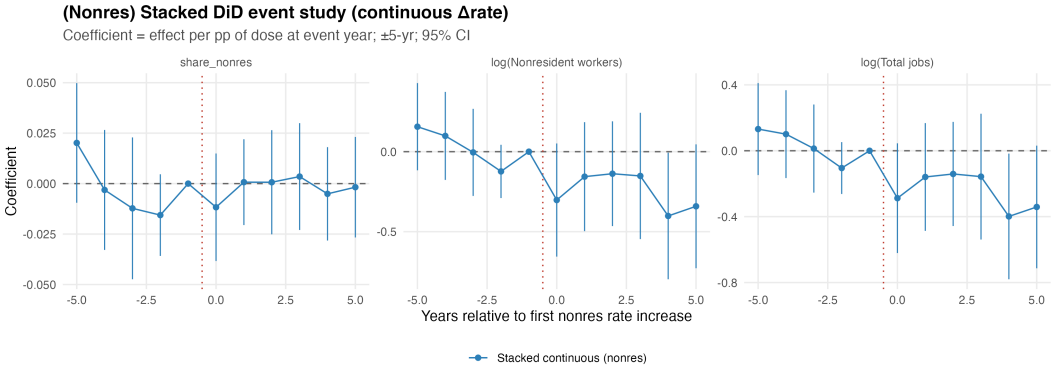
(4) Stacked DiD continuous $\Delta\tau$, nonres-side

$$Y_{ist} = \alpha_{is} + \gamma_{ts} + \beta\,\Delta\tau^{\text{nr}}_{ist} + \mathbf{X}'_{it}\boldsymbol{\phi} + \varepsilon_{ist}$$

	share_nonres	log(NonRes)	log(Total Jobs)
$\Delta\tau^{\text{nr}}$ (per pp)	0.000 (0.013)	−0.261 (0.165)	−0.262 (0.160)
N	61,557	61,557	61,557

- Direction matches resident side (negative); magnitude about 3× larger than resident-side stacked (−0.09).
- p ≈ 0.11. Underpowered: only 88 nonres-Inc munis and small dose variation (most muni nonres rates barely change).
- log(NonRes) and log(Total Jobs) move in tandem — consistent with nonresident workers being a large share of total jobs in many munis.

(4) Stacked continuous ES, nonres-side



Block B triangulation: Workplace × nonres rate

Estimator	share_nonres	log(NonRes)	log(Total Jobs)
(1) Single-event TWFE	−0.004	−0.289*	−0.286*
(2) DCDH ATE	−0.004	−0.298	−0.294
(3) Stacked binary	+0.000	−0.129	−0.130
(4) Stacked continuous	+0.000	**−0.261**	**−0.262**

- Direction robust: negative on level outcomes across all clean estimators.
- No estimator crosses 5%: nonres-side is underpowered (smaller treated group, smaller dose variation).
- share_nonres null everywhere — mechanically protected because numerator and denominator move together.
- Magnitudes are ∼2–3× resident-side coefficients — nonresident workers are more responsive per pp of tax than residents, in the data and consistent with the piecewise mechanism (residents face the home stack and a larger move cost; nonresidents can switch jobs).

Block C (Residence × home rate): motivation

So far, two experiments on the **workplace panel** (unit = work cousub):

- Workplace × **resident rate**: outcomes {share_live_in, log(Resident workers), log(Total Jobs)}.
- Workplace × **nonres rate**: outcomes {share_nonres, log(Nonres workers), log(Total Jobs)}.

Both index by where workers *work*. They miss residential moves when the worker keeps the same job: such a worker exits log(Resident workers) but the move itself is invisible.

Third experiment: **residence panel** (unit = home cousub), treatment = home muni's resident rate.

- residents_employed: total employed people *living* in B, any workplace.
- residents_work_outside: of those, the count commuting out.
- share_work_outside = residents_work_outside / residents_employed.

Aligned design: treatment (home rate) and unit (home muni) match. Cleanest direct test of residential sorting.

(1) Single-event TWFE, residence-side

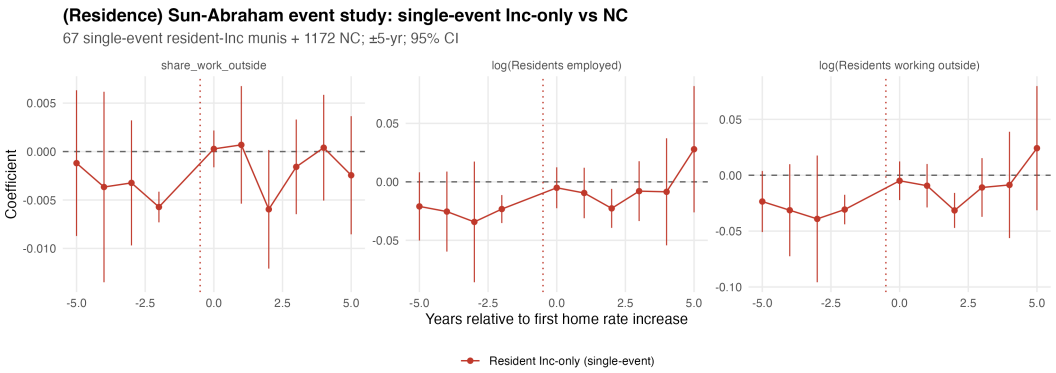
Sample: 1,172 NC + 67 single-event resident-Inc home munis (22 multi-event Inc dropped; 15 Dec, 54 flip-floppers excluded).

$$Y_{Bt} = \alpha_B + \gamma_t + \beta\, \tau^{\text{res}}_{Bt} + \mathbf{X}'_{Bt}\boldsymbol{\phi} + \varepsilon_{Bt}$$

	share_work_outside	log(Residents)	log(WorkOutside)
τ^{res}_{Bt} (home)	$+0.026^{***}$	-0.087^{**}	-0.054
	(0.008)	(0.038)	(0.045)

- 1pp ↑ in home rate → ∼9% drop in employed residents and +2.6pp in share commuting out.
- Absolute count of out-commuters doesn't move: residents leave faster than they switch to commuting.

(1) Single-event ES, residence-side (Sun-Abraham)



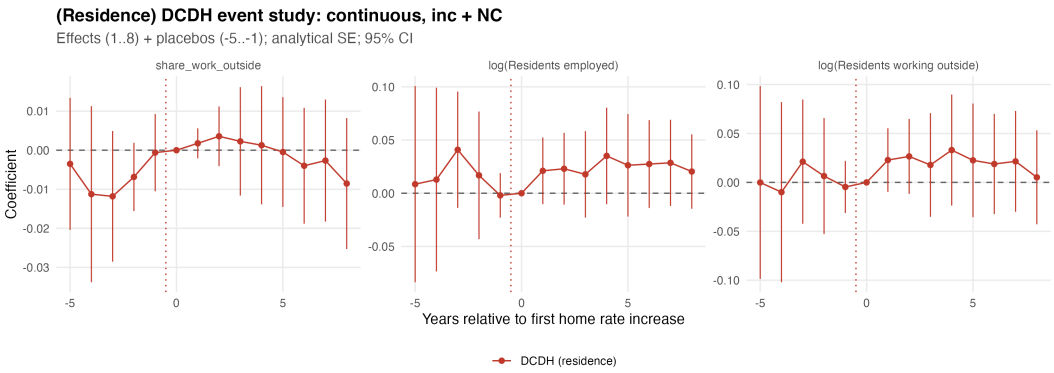
(2) DCDH, RESIDENCE-SIDE: CONTINUOUS, MULTI-EVENT

Inc (n_inc ≥ 1) + NC; analytical SE.

	share_work_outside	log(Residents)	log(WorkOutside)
DCDH ATE	−0.001	+0.075	+0.066
SE	(0.017)	(0.048)	(0.060)

- All three null with CIs spanning zero.
- Underpowered without bootstrap; analytical SEs are liberal on continuous DCDH per the authors.

(2) DCDH ES, RESIDENCE-SIDE



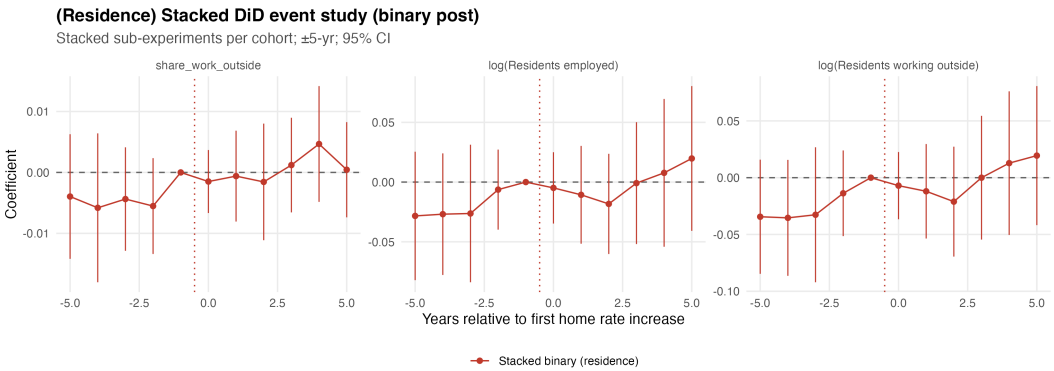
(3) Stacked DiD binary, residence-side

Stacked sub-experiments on single-event Inc + NC; ±5 yr.

	share_work_outside	log(Residents)	log(WorkOutside)
Treat × Post	+0.003	+0.010	+0.014
SE	(0.005)	(0.027)	(0.027)

- Binary post collapses dose; all null.

(3) Stacked binary ES, residence-side



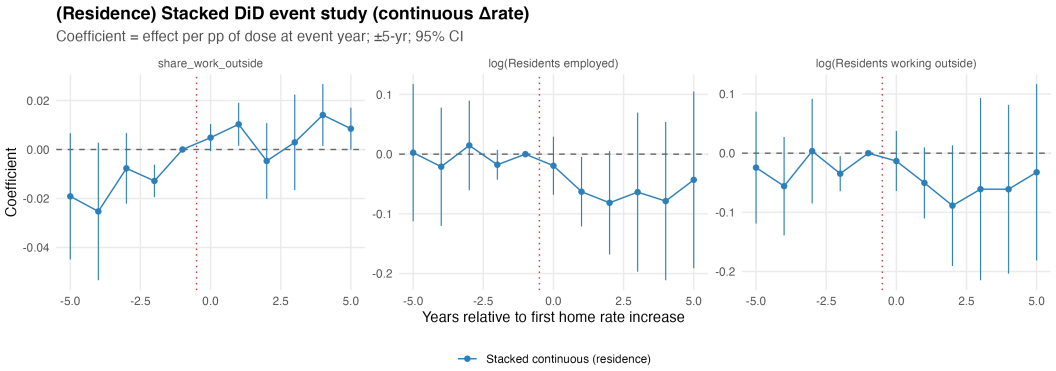
(4) Stacked DiD continuous $\Delta\tau$, residence-side

Treatment = $\Delta\tau_{Bt}^{\text{res}}$ vs pre-event mean.

	share_work_outside	log(Residents)	log(WorkOutside)
$\Delta\tau$	$+0.016^{*}$	-0.058	-0.037
SE	(0.009)	(0.040)	(0.049)
p	0.089	0.149	0.448

- share_work_outside marginally positive: residents reallocate toward out-commuting per pp of dose.
- log(Residents) trends negative (-0.058, $p = 0.15$); same sign as TWFE but not significant.

(4) Stacked continuous ES, residence-side



Block C triangulation: Residence × home rate

Estimator	share_work_out	log(Residents)	log(WorkOut)
(1) Single-event TWFE	$+0.026^{***}$	-0.087^{**}	-0.054
(2) DCDH ATE	-0.001	$+0.075$	$+0.066$
(3) Stacked binary	$+0.003$	$+0.010$	$+0.014$
(4) Stacked continuous	$+\mathbf{0.016}^{*}$	-0.058	-0.037

- TWFE and stacked-continuous agree in sign on share_work_outside (+) and log(Residents) (−); DCDH and stacked-binary are noisy nulls.
- Magnitude on log(Residents): −0.087 (TWFE) to −0.058 (stacked continuous) — about half the workplace-panel coefficient (−0.20 to −0.09) on log(Resident workers).
- Reading: residence panel isolates the residential-move channel; the gap with workplace-side is the live-and-work intersection collapsing from the work side (job changes, not moves).
- Sorting story holds, but smaller and noisier than the workplace-side estimate suggested.

THREE BLOCKS AT A GLANCE: STACKED-CONTINUOUS HEADLINE

Block	Question	Outcome	β per pp
A: Workplace panel × resident rate			
	Do live-and-work workers leave?	log(Resident wkrs)	$-$**0.091****
	Does total local employment fall?	log(Total Jobs)	$-$**0.072****
	Does the live-in share fall?	share_live_in	$-$0.005
B: Workplace panel × nonresident rate			
	Do nonresident workers flee?	log(NonRes wkrs)	$-$0.261
	Does total local employment fall?	log(Total Jobs)	$-$0.262
	Does the nonres share rise?	share_nonres	$+$0.000
C: Residence panel × home rate			
	Do residents leave the muni?	log(Residents emp)	$-$0.058
	Do residents shift to commuting out?	log(WorkOutside)	$-$0.037
	Does the share commuting out rise?	share_work_outside	$+$**0.016***

- Stacked-continuous $\Delta\tau$ on each block's Inc + NC sample. Single-event TWFE coefficients are larger (see each block's tri).
- A pins headline; C corroborates residential margin; B direction-consistent, not at 5%.

Is there a causal effect? Five independent reads

Yes on the resident-side. Five reads, different identification, same sign:

Evidence	Unit	Outcome	Result
Block A (4 estimators)	work cousub	log(Resident wkrs)	-0.04 to -0.20, sign robust
Block C (residence panel)	home cousub	log(Residents emp)	-0.087^{**} TWFE; -0.058 stacked
Act 32 break	work cousub	log(Resident wkrs)	-0.139^{***} pre $\rightarrow$ -0.209^{***} post
Piecewise (pair flows)	A$\times$B pair	log(flow)	-0.056^{***} when home flips to bind
Asymmetry (Q3 clean)	work cousub	share_live_in	Inc -0.016^{**}; Dec too thin

Weaker:

- DCDH (analytical SE) and stacked-binary often null; CIs span zero.
- Block B: direction-consistent, no estimator $< 5\%$ (88 nonres-Inc munis).
- `share_live_in` fragile; level outcomes are the robust headline.
- Dec (15 munis) too thin to test reversibility directly.

What do tax increases do? (1/2) Magnitudes

Headline (stacked continuous, Block A). 1pp ↑ in muni resident rate ⇒ ∼ 9% ↓ in resident workers, ∼ 7% ↓ in total jobs.

Decomposing the workplace-side decline:

$$\underbrace{-0.20}_{\text{Block A TWFE}} = \underbrace{-0.087}_{\text{Block C: residential moves}} + \underbrace{-0.11}_{\text{job-side adjustment within muni}}$$

- Roughly half is residential moves out of the taxing muni (Block C).
- Other half is the live-and-work intersection collapsing from the work side — workers staying put but losing/switching local jobs.
- Among residents who stay, share commuting out rises +2.6pp per pp of home rate (Block C TWFE).

What do tax increases do? (2/2) Mechanism

- **Total-jobs channel: consistent with local-services demand spillover, not direct nonresident taxation.** 74% of muni-rate hikes leave the nonres rate unchanged, and Block B (nonres rate) doesn't cross 5%. Industry-level decomposition pending (issue #2) to nail the demand mechanism.
- **SD rate doesn't sort.** Prior $+0.147^{**}$ on log(Resident) was a tract-centroid artifact; null under block-level LODES.
- **Enforcement-contingent.** Same statutory rates pre-Act-32 produce no effect. Consolidated withholding (post-2012) is what makes the rate bind. Collection capacity, not headline rate, is the policy lever.
- **Asymmetric.** Increases push residents out; decreases don't reverse (15 Dec munis too thin to identify). Consistent with move-cost / sticky residential decisions.

Summary of Findings

1. **Causal effect on residential sorting: yes.** Triangulated across 3 (panel × rate) blocks, 5 estimators, and the Act 32 break. Magnitude on log(Resident wkrs) ranges −0.05 to −0.20. Headline: -0.091^{**} (stacked continuous).
2. **Decomposition.** ∼Half the workplace decline is residential moves (Block C: -0.087^{**}); the other half is job-side adjustment. Share commuting out: +2.6pp per pp.
3. **Mechanism.** Resident rate ↑ → residents move out → local-services demand ↓ → total jobs ↓. Not the nonres rate (Block B), not SD (block-level LODES rules out the prior artifact).
4. **Enforcement-contingent.** Pre-Act-32 null on share_live_in, half-magnitude on log(Resident). Withholding capacity is the policy lever, not the statutory rate.
5. **Asymmetric.** Increases identified; decreases (15 munis) too thin. Consistent with move-cost story.
6. **What's not happening.** No SD sorting; no detectable nonresident-side response at 5%.

Next Steps

- Pair-level piecewise test on OD flows.
- Welfare: combine behavioral elasticity with revenue elasticity to bound efficiency cost of local income tax.

Thank You

Questions?

hocke069@umn.edu
github.com/MatthewHockert/local-income-tax-migration